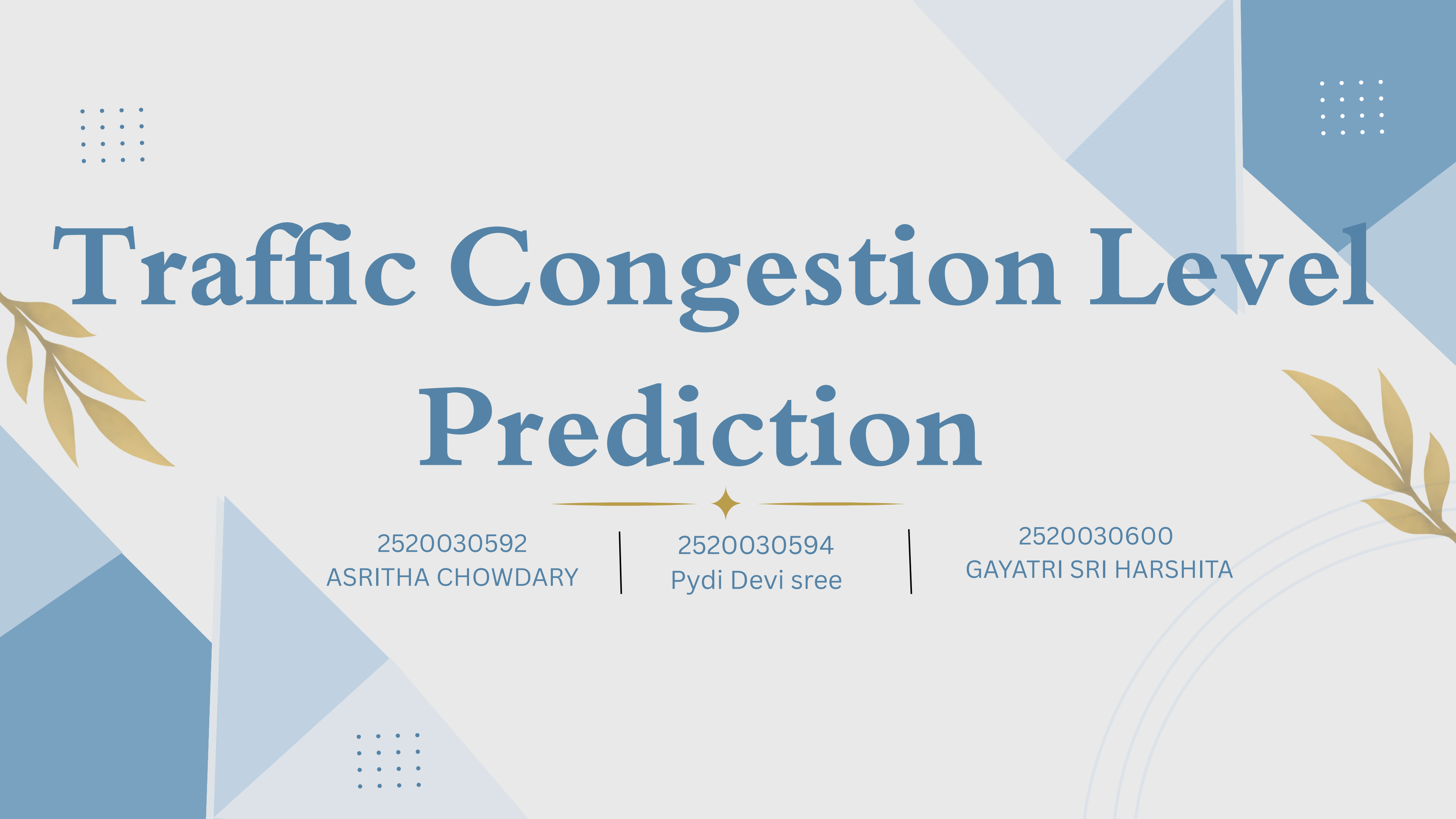




Traffic Congestion Level Prediction

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Abstract

Traffic congestion is a common problem in urban areas that causes travel delays, increased fuel consumption, and inconvenience to commuters. This project focuses on predicting traffic congestion levels using Machine Learning techniques. Historical traffic data containing factors such as vehicle count, average speed, traffic density, weather conditions, and time of day is analyzed and preprocessed. Two machine learning algorithms, Random Forest and Gradient-Boosted Trees, are implemented to classify traffic congestion into different levels such as Low, Medium, and High. The performance of both models is evaluated using suitable metrics such as accuracy, precision, recall, and F1-score. The models are then compared to identify the better-performing approach. The proposed system can help understand traffic patterns and support better traffic management and planning.



- Traffic congestion is a common problem in cities.
- It causes travel delays, fuel consumption, and inconvenience.
- Traffic conditions depend on several factors such as vehicle count, speed, weather, and time.
- Machine Learning can learn patterns from previous traffic data.
- Our project aims to predict congestion levels before or during traffic situations.



Data set using

~Interstat Traffic Volume Dataset

It contains information:

- Traffic volume
- Speed
- Holiday
- Temperature
- Weather
- Date and Time

~Congestion Level

we convert the traffic volume into three congestion levels.

- Hight
- Medium
- Low



Algorithms used

Random Forest

Gradient-Boosted Trees



Thank You

